

CHUAN MENG

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RESEARCH INTERESTS

I have been actively researching two core areas in AI: Natural Language Processing (NLP) and Information Retrieval (IR).

Specifically, my work focuses on leveraging large language models (LLMs) to build **information access agents** (a.k.a. deep research systems).

EDUCATION

University of Amsterdam, The Netherlands

October 2021 - June 2025

Ph.D. in Artificial Intelligence

Supervisors: Prof. dr. [Maarten de Rijke](#) and dr. [Mohammad Aliannejadi](#)

PhD thesis: Optimizing Agentic Workflows for Information Access [\[pdf\]](#)

Shandong University, China

September 2018 - June 2021

Master in Computer Science and Technology

Supervisors: Prof. dr. [Zhumin Chen](#), dr. [Pengjie Ren](#), and dr. [Zhaochun Ren](#)

Master's thesis: Research on Knowledge-Grounded Non-Task-Oriented Conversational System [\[pdf\]](#)

Shandong Normal University, China

September 2014 - June 2018

Bachelor in Electronic Commerce

WORK EXPERIENCE

The University of Edinburgh, United Kingdom

September 2025 - present

Postdoctoral Researcher (Research Associate), focusing on agentic information access

Funded by the Turing AI Fellowships

Supervisor: Dr. [Jeff Dalton](#) and Prof. [Mirella Lapata](#)

Amazon, United Kingdom

August 2024 - January 2025

Applied Scientist Intern, focusing on LLM-powered proactive conversational agents

Manager: Dr. [Gabriella Kazai](#). Mentor: Dr. [Francesco Tonolini](#).

PUBLICATIONS

As of May 2026, I have 700+ citations (Google Scholar) with an H-index of 16. I have authored papers published in proceedings/journals, such as ACL, EMNLP, NAACL, SIGIR, CIKM, AAAI, ECIR, and TOIS.

1. Revisiting Text Ranking in Deep Research

Chuan Meng, Litu Ou, Sean MacAvaney, and Jeff Dalton

SIGIR 2026: The 49th International ACM SIGIR Conference on Research and Development in Information Retrieval

[pdf](#), [code](#)

2. Re-Rankers as Relevance Judges

Chuan Meng, Jiqun Liu, Mohammad Aliannejadi, Fengran Mo, Jeff Dalton, and Maarten de Rijke

Under review

[pdf](#), [code](#)

3. **Generative Retrieval with Few-shot Indexing**
Arian Askari*, **Chuan Meng*** (co-first author), Mohammad Aliannejadi, Zhaochun Ren, Evangelos Kanoulas, and Suzan Verberne
ECIR 2026: The 48th European Conference on Information Retrieval
[pdf](#)
4. **Towards Adaptive Personalized Conversational Information Retrieval**
Fengran Mo, Yuchen Hui, Yuxing Tian, Zhaoxuan Tan, **Chuan Meng**, Zhan Su, Kaiyu Huang, Jian-Yun Nie
CIKM 2025: The 34th ACM International conference on Information and Knowledge Management
[pdf](#)
5. **UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations**
Fengran Mo, Yifan Gao, **Chuan Meng**, Xin Liu, Zhuofeng Wu, Kelong Mao, Zhengyang Wang, Pei Chen, Zheng Li, Xian Li, Bing Yin, Meng Jiang
ACL 2025: The 63rd Annual Meeting of the Association for Computational Linguistics
[pdf](#)
6. **Bridging the Gap: From Ad-hoc to Proactive Search in Conversations**
Chuan Meng, Francesco Tonolini, Fengran Mo, Nikolaos Aletras, Emine Yilmaz, and Gabriella Kazai
SIGIR 2025: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#), [code](#)
7. **Query Performance Prediction using Relevance Judgments Generated by Large Language Models**
Chuan Meng, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke
TOIS: ACM Transactions on Information Systems
[pdf](#), [code](#)
8. **SOLID: Self-seeding and Multi-intent Self-instructing LLMs for Generating Intent-aware Information-Seeking Dialogs**
Arian Askari, Roxana Petcu, **Chuan Meng**, Mohammad Aliannejadi, Amin Abolghasemi, Evangelos Kanoulas, and Suzan Verberne.
NAACL 2025: The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics
[pdf](#), [code](#)
9. **Zero-Shot and Efficient Clarification Need Prediction in Conversational Search**
Lili Lu, **Chuan Meng**, Federico Ravenda, Mohammad Aliannejadi and Fabio Crestani
ECIR 2025: The 47th European Conference on Information Retrieval
[pdf](#), [code](#)
10. **Improving the Re-Usability of Conversational Search Test Collections**
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi and Mohammad Aliannejadi
ECIR 2025: The 47th European Conference on Information Retrieval
[pdf](#), [code](#)
11. **Can We Use Large Language Models to Fill Relevance Judgment Holes?**
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi, Mohammad Aliannejadi
LLM4Eval: The First Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval
[pdf](#), [code](#)
12. **Ranked List Truncation for Large Language Model-based Re-Ranking**
Chuan Meng, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke.
SIGIR 2024: The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#), [code](#)

13. **Query Performance Prediction for Conversational Search and Beyond**
Chuan Meng
SIGIR 2024: The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#)
14. **LLM-based Retrieval and Generation Pipelines for TREC Interactive Knowledge Assistance Track (iKAT) 2023.**
Zahra Abbasiantaeb, **Chuan Meng**, David Rau, Antonis Krasakis, Hossein A. Rahmani, and Mohammad Aliannejadi.
TREC 2023: The Thirty-Second Text REtrieval Conference (**Our submitted runs ranked 1st**)
[pdf](#)
15. **Expand, Highlight, Generate: RL-driven Document Generation for Passage Reranking**
Arian Askari, Mohammad Aliannejadi, **Chuan Meng**, Evangelos Kanoulas, and Suzan Verberne
EMNLP 2023: The 2023 Conference on Empirical Methods in Natural Language Processing
[pdf](#), [code](#)
16. **System Initiative Prediction for Multi-turn Conversational Information Seeking**
Chuan Meng, Mohammad Aliannejadi, and Maarten de Rijke
CIKM 2023: The 32nd ACM International Conference on Information and Knowledge Management
[pdf](#), [code](#)
17. **Query Performance Prediction: From Ad-hoc to Conversational Search**
Chuan Meng, Negar Arabzadeh, Mohammad Aliannejadi, and Maarten de Rijke
SIGIR 2023: The 46th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#), [code](#)
18. **Performance Prediction for Conversational Search Using Perplexities of Query Rewrites**
Chuan Meng, Mohammad Aliannejadi, and Maarten de Rijke
QPP++ 2023: Query Performance Prediction and Its Evaluation in New Tasks Workshop co-located with The 45th European Conference on Information Retrieval
[pdf](#), [code](#)
19. **Initiative-Aware Self-Supervised Learning for Knowledge-Grounded Conversations**
Chuan Meng, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Tengxiao Xi, and Maarten de Rijke
SIGIR 2021: The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#)
20. **Conversations Powered by Cross-Lingual Knowledge**
Weiwei Sun^{*}, **Chuan Meng**^{*} (co-first author), Qi Meng, Zhaochun Ren, Pengjie Ren, Zhumin Chen, and Maarten de Rijke
SIGIR 2021: The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#), [code](#)
21. **DukeNet: A Dual Knowledge Interaction Network for Knowledge-Grounded Conversation**
Chuan Meng, Pengjie Ren, Zhumin Chen, Weiwei Sun, Zhaochun Ren, Zhaopeng Tu, and Maarten de Rijke
SIGIR 2020: The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval
[pdf](#), [code](#)
22. **RefNet: A Reference-aware Network for Background Based Conversation**
Chuan Meng, Pengjie Ren, Zhumin Chen, Christof Monz, Jun Ma, and Maarten de Rijke
AAAI 2020: The Thirty-Fourth AAAI Conference on Artificial Intelligence.
[pdf](#), [code](#)

ACADEMIC SERVICE

- arXiv, Moderator (cs.IR), February 2026–present
- Workshop organization:
 - **QPP++2025: Query Performance Prediction and its Applications in the Era of Large Language Models**
Chuan Meng, Guglielmo Faggioli, Mohammad Aliannejadi, Nicola Ferro, and Josiane Mothe
ECIR 2025: The 47th European Conference on Information Retrieval
6th–10th April 2025, Lucca, Italy
[pdf](#), [website](#)
- Program committee member: ACL 2023, EMNLP 2023–2021, SIGIR 2026–2024, COLING 2025, 2022 & 2020, CIKM 2025–2022, The Web Conf 2025–2024, WSDM 2025–2022, ECIR 2025–2024, ICTIR 2025 & 2023, SIGKDD 2022, ECML/PKDD 2022–2021, AAAI 2021
- ACL Rolling Review (ARR) reviewer 2025
- Journal reviewer: Transactions on Information Systems (TOIS), Information Processing and Management (IP&M)

TEACHING & SUPERVISION

- Tutorial organization:
 - **Conversational Search: Foundations, Large Language Models, and Agents**
Chuan Meng, Fengran Mo, Mohammad Aliannejadi, Jeff Dalton, and Jian-Yun Nie
ECIR 2026: The 48th European Conference on Information Retrieval
29 March to 2 April 2026, Delft, The Netherlands
[pdf](#), [slides](#), [website](#)
 - **Conversational Search: From Fundamentals to Frontiers in the LLM Era**
Fengran Mo, **Chuan Meng**, Mohammad Aliannejadi, and Jian-Yun Nie
SIGIR 2025: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval
13th–17th July 2025, Padua, Italy
[pdf](#), [slides](#), [website](#)
 - **Query Performance Prediction: Theory, Techniques and Applications**
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri
WSDM 2025: The 18th ACM International Conference on Web Search and Data Mining
10th–14th March 2025, Hanover, Germany
[pdf](#), [slides](#)
 - **Query Performance Prediction: Techniques and Applications in Modern Information Retrieval**
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri
SIGIR-AP 2024: The 2nd ACM SIGIR-AP conference
9th–12th December 2024, Tokyo, Japan
[pdf](#), [slides](#)
 - **Query Performance Prediction: From Fundamentals to Advanced Techniques**
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri
ECIR 2024: The 46th European Conference on Information Retrieval
24th–28th March 2024, Glasgow, UK
[pdf](#), [slides](#)
- Lecturer
 - **Course**: Recommender Systems (about **50** students)
Lecture: Sequential and Large Language Model-based Recommender Systems
Institution: University of Amsterdam
Year: June 2025
My responsibilities: Course design and lecture delivery

[slides](#)

- **Guest Lecture:** Conversational Search: From Fundamentals to Frontiers in the LLM Era (about 40 undergraduate students)

Institution: University of Massachusetts Amherst (UMass Amherst)

Host: [Razieh Negin Rahimi](#)

Year: October 2025

[slides](#)

- Teaching assistant
 - **Course:** Information Retrieval (about 60 students)
Institution: University of Amsterdam
Year: 2023
My responsibilities: project design and grading
 - **Course:** Information Retrieval (about 60 students)
Institution: University of Amsterdam
Year: 2022
My responsibilities: project design and grading
 - **Course:** Information Retrieval (about 60 students)
Institution: Shandong University
Year: 2020
My responsibilities: assignment design, project design and grading
 - **Course:** Natural Language Processing (about 70 students)
Institution: Shandong University
Year: 2019
My responsibilities: assignment design, project design and grading
- PhD mentorship (Research-oriented): [Lili Lu](#), Università della Svizzera italiana (USI), with Prof. dr. [Fabio Crestani](#)

INVITED TALKS

- **Opportunities and Challenges of LLMs in Information Retrieval**
21 October 2024, University of Glasgow
Host: Iadh Ounis, professor at the University of Glasgow
[slides](#), [post](#)
- **Predicting the Right Moment for System Initiative in Mixed-Initiative Conversational Search**
26 August 2024, Amazon (London)
Host: Gabriella Kazai, principal applied scientist at Amazon
[slides](#)
- **Opportunities and Challenges of LLMs in Information Retrieval**
14 August 2024, Objective, Inc.
Host: Pablo Mendes, co-founder & CEO at Objective, Inc.
[slides](#)
- **Opportunities and Challenges of LLMs in Information Retrieval**
17 April 2024, Amazon (Seattle)
Host: Shervin Malmasi, applied science manager at Amazon
[slides](#)
- **System Initiative Prediction and Query Performance Prediction for Conversational Information Seeking**
3 Nov 2023, University College London (UCL)
Host: Xi Wang, lecturer at the University of Sheffield
[slides](#), [post](#)
- **Query Performance Prediction for Conversational Search**
18 May 2023, University of Glasgow

Host: Iadh Ounis, professor at the University of Glasgow
[slides](#), [post](#)

ADMINISTRATION ACTIVITIES

- Manager for the [IRLab LinkedIn](#) account, 2025
- Webmaster for the [IRLab website](#), 2023–2024
- Chair for internal seminars at IRLab, 2023
- Lead organizer of the IRLab BBQ event, 2023

RESEARCH FUNDING PROPOSALS

- **Amazon Research Award** (submitted)
Co-Principal Investigator (lead proposal)
Developing agentic AI systems for clinical decision support
- **National Key Research and Development Program of China** (Project No.: 2020YFB1406700)
Contributor (methodology design)
Development of an AI-assisted writing robot system based on news reporting scenarios
- **National Natural Science Foundation of China (NSFC)** (Project No.: 61972234)
Contributor (methodology design)
Research on Explainable Medical Health Question-Answering Technology Based on Multi-Source Heterogeneous Data Fusion
- **National Natural Science Foundation of China (NSFC)** (Project No.: 62072279)
Contributor (methodology design)
Research on Conversational Recommendation Systems and Functional Enhancements Based Primarily on Implicit Feedback Data

SCHOLARSHIPS & AWARDS

- Excellent Master's Thesis of Shandong Province, 2022
- National Scholarship (China), 2020/2016
- Outstanding Graduates of Shandong Province, 2021/2017
- SIGIR Student Travel Grant, 2020
- Scholarship for Outstanding Postgraduate Cadres of Shandong University, 2020/2019
- AAAI Student Scholar Scholarship, 2019
- Academic Scholarship for Master Students of Shandong University, 2019
- Outstanding Students of Shandong Province, 2017

RESOURCES

I have curated the following resources:

- a [code repository](#) (**over 2,500 visitors**) providing scripts for fine-tuning open-source LLMs to generate relevance judgments, within a Python/PyTorch framework.
- a [code repository](#) (**over 3,600 visitors**) providing a comprehensive implementation of query performance prediction (QPP) methods, within a unified Python/PyTorch framework.
- a [code repository](#) (**over 1,400 visitors**) offering a comprehensive implementation of ranked list truncation methods, within a unified Python/PyTorch framework.
- a [paper reading list](#) (**over 260 stars**) on knowledge-grounded dialogue systems.

Also, I have contributed to key open-source toolkits in information retrieval, including:

- [Pyserini \(v0.19.2\)](#), a Python toolkit for reproducible information retrieval research.
- [Tevatron](#), a flexible and efficient Python toolkit for training and inference of neural retrieval models.

REFERENCES

Dr. [Jeff Dalton](#)

Chancellor's Fellow (Associate Professor), Turing AI Acceleration Fellow
The University of Edinburgh
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Prof. dr. [Maarten de Rijke](#)

Distinguished University Professor
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Dr. [Mohammad Aliannejadi](#)

Assistant Professor
University of Amsterdam
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Dr. [Zhaochun Ren](#)

Associate Professor
Leiden University
z.ren@liacs.leidenuniv.nl

Prof. dr. [Fabio Crestani](#)

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